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Distributed Portfolio Optimization & Decentralized Pricing

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Abstract

We present a distributed portfolio construction framework based in network structures and combinatorial optimization. Unlike traditional centralized methods, our approach decomposes the portfolio under consideration into overlapping sub-portfolios, each reflecting a thematic strategy or mandate. This decomposition is guided by the powerset of the available set of assets, capturing all meaningful groupings and inducing a hypergraph structure where assets appear in multiple sub-portfolios. We solve the resulting distributed portfolio optimization problem using a primal-dual algorithm: primal variables represent sub-portfolio allocations, while dual variables emerge as shadow prices on coupling assets, offering a natural pricing interpretation and a link to microeconomic theory (general equilibrium). The framework integrates both internal portfolio signals and external models such as CAPM. We illustrate the methodology using GICS sector ETFs, constructing sub-portfolios aligned with macroeconomic themes.

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1 Introduction

Optimization in finance refers to the process of constructing mathematical models that align with financial objectives while incorporating domain-specific constraints. These settings include portfolio construction, trading execution, and risk budgeting. For a comprehensive survey of optimization methods in finance, see Cornuejols et al. (2018). In this work, we focus on portfolio optimization—the problem of selecting the optimal distribution of assets to achieve desired trade-offs between return and risk.

Modern portfolio theory, initiated by Markowitz in Markowitz (1959), casts this as a mean-variance optimization problem: maximizing expected returns while minimizing return variance, resulting in a quadratic programming formulation. Traditional methods, including extensions summarized in Boyd et al. (2024), often assume a centralized optimization process, treating assets as a flat list and assigning global weights. These methods overlook the structured, multi-layered decision processes typical in institutional asset management.

We introduce a novel methodology: a distributed portfolio construction framework based on overlapping sub-portfolios. Each sub-portfolio represents a localized investment strategy, macroeconomic theme, or thematic mandate. Overlapping exposures across sub-portfolios create consistency constraints—ensuring that an asset's total allocation remains coherent across all portfolios in which it appears. This setting gives rise to a coupled optimization problem, naturally modeled by network structures.

It is important to note the equivalence between decentralized and centralized mean-variance optimization models. Despite structural differences, both approaches are informationally equivalent, using the same input data and operating in the same asset space. The decentralized formulation decomposes the problem into sub-portfolios connected by consensus constraints, maintaining the economic intent of the centralized model while offering modular risk control.

The use of network theory in financial applications is well-documented, including studies of systemic risk Allen and Babus (2009), contagion during crises Elliott et al. (2014), supply chains Nagurney and Siokos (2012), and interbank networks Babus (2016), Demirer et al. (2018). Recent work such as Ricca and Scozzari (2024) and Wang et al. (2024) has applied network-based methods to portfolio construction, laying groundwork for our approach.

Here, we adopt a hypergraph-based formulation, where each node represents an asset and each hyperedge corresponds to the common value of the overlapping assets. Our construction leverages the power set of the assets under consideration, from which we select meaningful subsets (sub-portfolios) based on economic relevance. We emphasize that while the full power set offers combinatorial completeness, only a curated selection of subsets is typically meaningful from a financial economics standpoint. For illustrative purposes, we demonstrate both the exhaustive case (3-asset universe) and a realistic example (GICS sectors) involving six economically motivated sub-portfolios.

The overlapping structure induces coupling variables—shared asset weights across sub-portfolios—which define the core interdependence of the system. To model this, we use a hypergraph representation Berge (1973), where more than two sub-portfolios may share a given asset. This structure supports distributed optimization techniques, as surveyed in Yang et al. (2019), with foundational roots in Tsitsiklis (1984). We

employ the hypergraph Laplacian Bolla (1993) to formulate the optimization problem, and solve it using a primal-dual algorithm.

In our setting, the primal variables represent allocations within each sub-portfolio. The dual variables, arising from consistency constraints on coupling variables, carry economic meaning—they act as shadow prices Kelly et al. (1998), enabling a pricing mechanism across sub-portfolios. This mechanism naturally aligns with ideas from general equilibrium theory, linking decentralized allocators through market-clearing prices. Dual variables also allow for integration of external valuation models such as CAPM or arbitrage pricing, offering a unified internal-external pricing framework.

In the finance literature, such overlapping portfolio structures are common in practice. They may arise from customized services offered by financial providers, or organically through separate mandates with shared exposures. While this paper focuses on fully coupled cases—where all assets are shared across sub-portfolios—the framework can be extended to handle local (non-coupled) variables as well, following techniques such as those in Papastaikoudis et al. (2024).

As a practical example, we apply our methodology to the Global Industry Classification Standard (GICS) sector ETFs. The 11 sectors (e.g., XLK for Technology, XLF for Financials, etc.) are reorganized into six overlapping sub-portfolios based on macroeconomic themes. This setting yields a nontrivial hypergraph structure and serves as an ideal real-world demonstration.

Yet the generality of the framework extends beyond GICS portfolios. Potential applications include:

- Pension fund management, where asset class teams optimize locally under global constraints;
- ESG investing, where portfolios overlap due to shared environmental or social factors;
- Multi-manager hedge funds and fund-of-funds, where decentralized allocators operate semi-independently under firm-level risk and liquidity controls.

From an operational perspective, a financial institution (e.g., asset manager or bank) might offer a menu of overlapping sub-portfolios drawn from a common asset universe. Investors can express preferences across these sub-portfolios, and allocations across shared assets are priced internally via dual variables. These prices reflect the marginal value of including an asset in multiple themes/mandates—offering a financial service innovation. Overall the model offers a scalable, interpretable, and economically grounded framework for modern decentralized portfolio design.

Beyond its core contributions, the proposed framework opens several promising avenues for future exploration, including machine learning-guided subset selection, dynamic rebalancing strategies, behavioral finance integrations, and the potential tokenization of sub-portfolios for decentralized finance applications. These directions underscore the flexibility of the architecture and its adaptability to both traditional and emerging financial paradigms.

The structure of the paper is as follows: Section 2 formulates the hypergraph-based distributed portfolio optimization problem, introduces the primal-dual algorithm, and

interprets its pricing features. Section 3 provides a numerical example using a three-asset universe and its full power set. Section 4 presents the architectural superiority of the decentralized portfolio construction over its centralized counterpart. Finally, in section six we apply the framework to the GICS sector ETF universe and validate the model with real-world data. We also include a subsection outlining potential improvements and directions for future work. An appendix of proofs and the source code used for the data extraction is provided at the end of the paper.

2 Distributed Portfolio Optimization Problem

We are given a finite set of financial assets A, and construct K overlapping sub-portfolios, each associated with a subset from the power set $\mathcal{P}(A)$ of A, where the power set is the set of all possible subsets of a given set, including the empty set and the full set itself. We study the following distributed mean-variance portfolio optimization problem for the K different sub-portfolios:

$$\min_{x=[x_1,\ldots,x_K]} \sum_{i=1}^{K} a_i f_i(x_i)$$

$$\begin{aligned} \text{s.t.} \quad & x^T \cdot \mathbf{1} = 1 \quad \text{(budget constraint)} \\ & Lx = 0 \quad \text{(consistency across shared assets)} \\ & x \geq 0 \quad \text{(no short-selling and leveraging)} \end{aligned} \tag{1}$$

where

- x_i denotes the allocation vector for sub-portfolio i and has dimension p_i. It is a coupling variable, i.e., the vector components of x_i appear in other x_i's as well.
- Vector $x = [x_1, ..., x_K]^T$ is the stacked global decision vector of the optimization problem (1).
- a_i is the respective weight of the ith portfolio. The portfolio weights sum to one, i.e., $\sum_{i=1}^{N} a_i = 1$ and if not otherwise specified the weights are equally distributed, i.e., $a_i = \frac{1}{K}$. Parameters a_i's can also be interpreted as risk aversion measures for the respective sub-portfolios.
- $f_i(x_i)$ are quadratic functions of the form $f_i(x_i) = x_i^T Q_i x_i - x_i^T \mu_i + c_i$, where
  - Q_i: Covariance matrix of returns in sub-portfolio i (captures risk).
  - μ_i: Expected return vector for sub-portfolio i.
  - c_i: Fixed cost for managing sub-portfolio i.
- L is the Laplacian matrix of the hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ that represents the coupling interconnections among the assets of the different sub-portfolios. This enforces consistency in allocations across portfolios sharing the same asset.

2.1 Hypergraph Interpretation

We represent the structural interdependencies across sub-portfolios using a hypergraph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$. The node set $\mathcal{V}$ is partitioned as $\mathcal{V} = \{\mathcal{V}_1, \ldots, \mathcal{V}_K\}$, where each subset $\mathcal{V}_i$ corresponds to the variable components associated with sub-portfolio i, that is, the elements of the allocation vector x_i. The hyperedge set $\mathcal{E}$ encodes the coupling structure among the x_i's by capturing the common elements shared across multiple sub-portfolios. Each hyperedge links variables that represent the same underlying asset, thereby enforcing consistency across shared values.

The hyperedge set, $\mathcal{E}$ denotes all common values, with $|\mathcal{E}|$ representing its cardinality. These shared components are compactly expressed by a vector $z \in \mathbb{R}^{|\mathcal{E}|}$, where each entry corresponds to a unique shared allocation variable. As a result, the hypergraph $\mathcal{H}$ satisfies the following structural properties:

- The total number of nodes is $|\mathcal{V}| = p = p_1 + \cdots + p_K$.
- The number of hyperedges is $|\mathcal{E}| = |A|$, i.e., the number of the underlying assets (all of them are coupling).
- The vertex degree matrix is $D_V = I_{p\times p}$.
- The edge degree matrix is given by $D_E = \operatorname{diag}\left\{|\mathcal{E}_1|, \ldots, |\mathcal{E}_{|A|}|\right\}$,

where $\mathcal{E}_j$ denotes the set of nodes connected by the j-th hyperedge. The hypergraph incidence matrix E has dimensions $|\mathcal{V}| \times |\mathcal{E}|$ and is of the following structure,

$$E = \begin{bmatrix} E_1 \\ \vdots \\ E_K \end{bmatrix},$$

with each $E_i \in \mathbb{R}^{p_i \times |\mathcal{E}|}$ defined elementwise by:

$$E_i^{lj} = \begin{cases} 1, & \text{if } x_i^l = z_j, \\ 0, & \text{otherwise,} \end{cases} \quad \text{for all } 1 \le l \le p_i,\ 1 \le j \le |\mathcal{E}|. \tag{2}$$

For clarity, we use x_i^j to denote the j-th component of i-th sub-portfolio that is linked to the j-th shared value in z, for $i = 1, \ldots, K$ and $j = 1, \ldots, |\mathcal{E}|$. A key property of our construction is that $D_E = E^T E$. As a result, the hypergraph Laplacian takes the normalized form:

$$L = D_V - E D_E^{-1} E^T = I - E(E^T E)^{-1} E^T. \tag{3}$$

This Laplacian matrix encodes the structure of the coupling constraints and is central to enforcing alignment across the overlapping components of the sub-portfolios.

Interpretation of the Optimum

The optimal solution x^* of (1) represents the **allocation of capital across interconnected sub-portfolios**, satisfying global budget and consistency constraints.

Each vector x_i^* corresponds to an optimal internal allocation, while $Lx = 0$ ensures consistent investment in overlapping assets. The objective value contains:

- A **quadratic risk term** $x_i^T Q_i x_i$
- A **linear return term** $-x_i^T \mu_i$
- A **fixed cost** c_i

Thus, the optimum value reflects a **risk-adjusted performance metric**, capturing more than just risk and offering a richer objective.

Remark 1. *Although the power set of A has $2^{|A|}$ subsets, we use only structured clusters rather than all subsets. For $|A| = 1000$, selecting meaningful subsets (e.g., sectors or ESG themes) makes the method scalable and suitable for large portfolios. The problem remains convex and can be solved using distributed methods.*

2.2 Generating Financial Products from Subsets of Assets

One of the most compelling features of the proposed optimization framework is its ability to generate a large number of distinct financial products from a relatively small initial set of assets. Specifically, by considering combinations of subsets of assets from the original portfolio (i.e., the powerset), it becomes feasible to design a rich variety of customized financial products, each tailored to different investor preferences or market strategies.

For example, from just 10 different assets, one can theoretically construct $2^{10} = 1023$ unique subsets (excluding the empty set), each representing a different sub-portfolio. When paired with variable asset weightings and specific investment constraints (e.g., ESG themes, sector allocation limits, or regulatory filters), each sub-portfolio can be interpreted as a distinct financial product. These products differ not only in composition but also in their respective risk-return characteristics, making them suitable for a wide range of investment strategies—from conservative to aggressive.

This observation highlights the combinatorial power of the approach: starting with just 10 assets, a financial institution can structure over 1000 distinct financial offerings, each corresponding to a meaningful and implementable investment option. This enables hedge funds or asset managers to:

- Offer tailored products aligned with client-specific preferences;
- Rapidly prototype and evaluate new trading or allocation strategies;
- Segment products by risk appetite, regulatory needs, or ESG alignment;
- And efficiently diversify offerings without expanding the base asset universe.

Moreover, this idea scales naturally to larger asset universes, such as portfolios with 1000+ assets. By leveraging intelligent clustering and subset selection strategies—rather than enumerating the full power set—the model maintains its expressiveness and product diversity while remaining computationally efficient and practically implementable.

2.3 Primal-Dual Algorithm

We derive a primal-dual algorithm for solving the distributed portfolio optimization problem defined in (1). The Lagrangian function associated with this problem is given by:

$$\mathcal{L}(x, \lambda, v, k) = f(x) - \lambda^T \left(\sum_{i,j} x_i^j - 1 \right) - v^T L x - k^T x,$$

where $f(x) = \sum_{i=1}^{K} f_i(x_i)$ is the total objective function, $\lambda \in \mathbb{R}$ is the dual variable corresponding to the budget constraint $x^T \cdot \mathbf{1} = 1$, $v \in \mathbb{R}^p$ is the dual variable associated with the Laplacian constraint $Lx = 0$, and $k \in \mathbb{R}^p$ is the dual variable enforcing the non-negativity constraint $x \geq 0$. To compute a saddle point of the Lagrangian, we propose the following continuous-time primal-dual algorithm:

$$\dot{x} = -\nabla f(x) + Lv + \mathbf{1} \odot \lambda, \tag{4a}$$

$$\dot{\lambda} = 1 - \sum_{i,j} x_i^j, \quad \text{with projection: } x_i^j \leq 0 \Rightarrow x_i^j = 0, \text{ then return to (4a), (4b)}, \tag{4b}$$

$$\dot{v} = -Lx, \quad \text{with projection: } v_i^j \leq 0 \Rightarrow v_i^j = 0, \text{ then return to (4a), (4c)}. \tag{4c}$$

In this formulation:

- The operation $\odot$ denotes element-wise multiplication.
- $\mathbf{1}$ is a vector of ones conforming in dimension to x.

This dynamical system iteratively updates both the primal variable x and the dual variables (λ, v), incorporating projections to ensure feasibility with respect to the positivity and consensus constraints. The formulation is well-suited for decentralized or distributed optimization contexts, particularly when asset allocations are coupled through shared components and network structures.

We now formally establish the convergence properties of the proposed primal-dual dynamics under appropriate assumptions on the objective function and the constraint structure. The analysis includes the positivity constraint via projection using non linear control standard tools and invariant set theory.

Theorem 1 (Global Convergence of Primal-Dual Dynamics)**.** *The continuous-time primal-dual dynamics (4a)-(4c) converge globally to a saddle point* (x^*, λ^*, v^*) *of the Lagrangian*

$$\mathcal{L}(x, \lambda, v) = f(x) - \lambda(\mathbf{1}^\top x - 1) - v^\top L x,$$

which corresponds to the optimal solution of the original optimization problem (1).

Remark 2. *A complete proof of this theorem is provided in the appendix.*

Remark 3. *While Pareto frontiers are commonly used in multi-objective optimization to visualize trade-offs between competing objectives (e.g., return vs. risk), our setting differs fundamentally. The distributed portfolio construction problem we study is formulated as a convex optimization problem with a single, well-defined objective and*

explicit structural constraints. As a result, our primal-dual algorithm yields a unique optimal solution — including allocations, objective value, and dual prices — for a given input setup. Therefore, the concept of a Pareto frontier is not necessary or applicable here, unless the problem is explicitly reformulated to involve multiple conflicting objective functions. Our framework focuses instead on structural decomposition, equilibrium pricing, and interpretability under a unified economic objective.

2.4 Decentralized Pricing Mechanism for Coupled Assets

The dual variables in the proposed primal-dual optimization algorithm can be interpreted as *shadow prices* or *implied equilibrium prices* associated with the structural constraints in our portfolio model. This economic interpretation enables the design of a decentralized pricing mechanism for the newly introduced financial products that arise from shared exposures across sub-portfolios.

Let the vector $z \in \mathbb{R}^{|A|}$ represent the prices of $|A|$ underlying or primitive financial assets that give rise to the coupling constraints in the decision variables x_i. We define a pricing mechanism for the coupling components, denoted by $p^* \in \mathbb{R}^{|A|}$, using the asymptotic (equilibrium) values of the dual variables v^* corresponding to the Laplacian constraint $Lx = 0$. The mechanism is given by:

$$p^* = \left[\text{inv}(E^\top v^*) \odot v^* + \mathbf{1}\right] \odot z, \tag{5}$$

where:

- E is the incidence matrix of the hypergraph representing the asset-subportfolio couplings,
- $\text{inv}(\cdot)$ denotes element-wise inversion, i.e., $\text{inv}(x_i) = 1/x_i$ for $x_i \neq 0$,
- $\odot$ is the Hadamard (element-wise) product,
- $\mathbf{1}$ is a vector of ones of appropriate dimension.

Equation (5) expresses the price of each coupled component as a linear adjustment over the base asset price z_j, where the adjustment reflects the implied equilibrium loadings derived from the dual variables. More explicitly, consider the component x_i^j associated with asset z_j and involved in k different sub-portfolios. Its equilibrium price can be written as:

$$p_i^{j*} = \left(\frac{v_i^j}{\sum_{\ell=0}^{k} v_\ell^j} + 1 \right) \cdot z_j.$$

This expression implies that the price of x_i^j is a weighted markup over the base asset price z_j, with the markup proportional to the relative dual weight v_i^j within the pool of sub-portfolios sharing exposure to z_j. Economically, this reflects the notion that investors accessing x_i^j through this structured product face reduced idiosyncratic risk (via coupling and diversification), and hence, a premium is justified.

The stochastic version of the pricing mechanism can be naturally derived by applying the expectation operator to both sides of (5), yielding:

$$\mathbb{E}(p^*) = \left[\text{inv}(E^\top v^*) \odot v^* + \mathbf{1}\right] \odot \mathbb{E}(z), \tag{6}$$

indicating that the expected price of each component is directly governed by the expected price of the underlying assets. Thus, the model remains compatible with any classical asset pricing theory applied to z, including the Capital Asset Pricing Model (CAPM), arbitrage pricing theory (APT), or models with stochastic discount factors. This flexibility ensures that our pricing mechanism can be embedded within existing financial pricing frameworks, allowing seamless integration with standard valuation models.

3 Distributed Portfolio Design with Powerset-Based Sub-Portfolios

In this section, we present a comprehensive example that highlights the advantages of our decentralized financial product design under a hypergraph-based portfolio allocation framework. Beginning with a traditional centralized mean-variance optimization over a small asset universe of three securities, we extend the formulation to include partially overlapping and fully combinatorial sub-portfolios. These sub-portfolios are constructed using selected and exhaustive subsets of the asset powerset, respectively. By embedding these allocations within a distributed optimization problem and solving it using our proposed primal-dual algorithm, we observe notable improvements in portfolio efficiency, risk diversification, and pricing transparency. Furthermore, the economic interpretation of dual variables as endogenous shadow prices enables the emergence of a decentralized pricing mechanism. This mechanism links the premium of sub-portfolio instruments to their marginal contribution in risk-reduction — offering a market-consistent valuation approach for modular financial instruments.

3.1 Centralized Case

The standard mean variance optimization problem for a portfolio with three assets, $A = \{x^1, x^2, x^3\}$, is presented below,

$$\min_{x'=(x^1,x^2,x^3)} f(x)$$
$$s.t.\ x^T \cdot \mathbf{1} = 1,\ x^j \geq 0\ \forall j$$

where

$$f(x) = x^T Q x - x^T \mu + c$$

and by assuming that the parameters are

$$Q = \begin{pmatrix} 4.1294 & 1.4527 & 0.9273 \\ 1.4527 & 4.4150 & 1.1068 \\ 0.9273 & 1.1068 & 3.3435 \end{pmatrix}, \mu = (0.3, 0.25, 0.1), c = 1$$

we trivially find the asymptotic values,

$$x^* = (0.3217, 0.2567, 0.4217)$$

with the optimal value of the centralized optimization problem to be

$$f(x^*) = 2.8408.$$

For the pricing of the three assets standard methods can be used such as the CAPM model or the arbitrage pricing theory.

3.2 Specific Partitions Case

In this case assuming the same set of assets, A we use a subset Σ of the partitions formed by the powerset $\mathcal{P}(A)$. We have the following set of available sub-portfolios $\Sigma = \{\Sigma_1, \Sigma_2, \Sigma_3\}$ where they are visually described in Figure 1 with the respective distributed mean variance optimization problem being of the following form

$$\min_{[x_1,x_2,x_3]} \sum_{i=1}^{3} a_i f_i(x_i)$$

$$\begin{aligned} s.t. \quad & x = Ez \\ & x^T \cdot \mathbf{1} = 1,\ x_i^j \geq 0 \\ & a^T \ \cdot \mathbf{1} = 1,\ a_i \geq 0 \end{aligned}$$

where

$$\begin{aligned} f_i(x_i) &= x_i^T Q_i x_i - x_i^T \mu_i + c_i \\ x_1 &= (x_1^1, x_1^2, x_1^3) \\ x_2 &= (x_2^1, x_2^2) \\ x_3 &= (x_3^1, x_3^3) \end{aligned}$$

with x_i^j denoting the ith financial instrument belonging to the jth sub-portfolio, $\mathbf{1}$ is a vector of ones of appropriate dimensions, $a = (a_1, a_2, a_3)$ is a simplex of parameters that denotes the respective weights of the sub-portfolios and finally c_i represents a fixed cost for choosing ith sub-portfolio. Finally, we have that the variance-covariance matrices and the respective mean value vectors are provided below.

$$Q_1 = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} \\ \sigma_{2,1} & \sigma_2^2 & \sigma_{2,3} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_3^2 \end{pmatrix}, \mu_1 = (\mu_1, \mu_2, \mu_3)$$

$$Q_2 = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_2^2 \end{pmatrix}, \mu_2 = (\mu_1, \mu_2)$$

$$Q_3 = \begin{pmatrix} \sigma_1^2 & \sigma_{1,3} \\ \sigma_{3,1} & \sigma_3^2 \end{pmatrix}, \mu_3 = (\mu_1, \mu_3).$$

The hypergraph incidence matrix E and its respective Laplacian matrix are given below:

$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, L = \begin{pmatrix} \frac{2}{3} & 0 & 0 & \frac{-1}{3} & 0 & \frac{-1}{3} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{-1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & \frac{-1}{2} \\ \frac{-1}{3} & 0 & 0 & \frac{2}{3} & 0 & \frac{-1}{3} & 0 \\ 0 & \frac{-1}{2} & 0 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{-1}{3} & 0 & 0 & \frac{-1}{3} & 0 & \frac{2}{3} & 0 \\ 0 & 0 & \frac{-1}{2} & 0 & 0 & 0 & \frac{1}{2} \end{pmatrix}$$

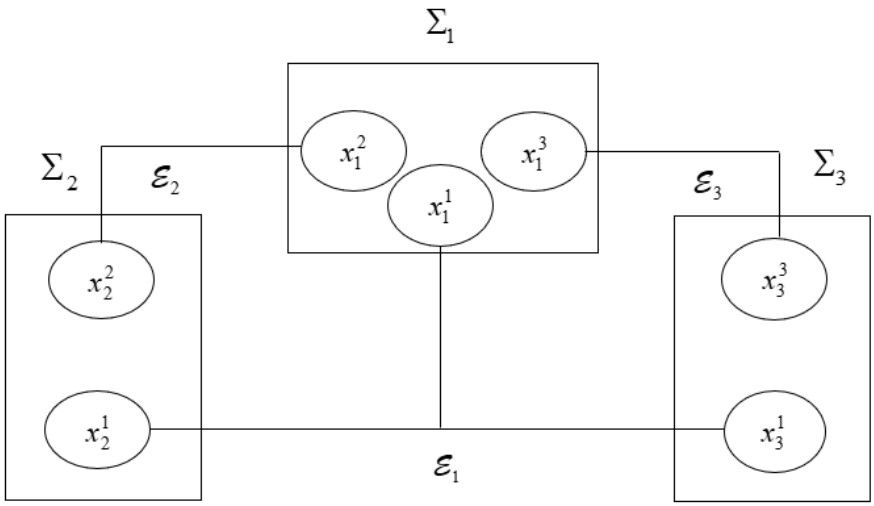


Fig. 1 Specific Portfolio Overlapping

From the centralized case we have that the variance-covariance matrices are

$$Q_1 = \begin{pmatrix} 4.1294 & 1.4527 & 0.9273 \\ 1.4527 & 4.4150 & 1.1068 \\ 0.9273 & 1.1068 & 3.3435 \end{pmatrix}, Q_2 = \begin{pmatrix} 4.1294 & 1.4527 \\ 1.4527 & 4.4150 \end{pmatrix}, Q_3 = \begin{pmatrix} 4.1294 & 0.9273 \\ 0.9273 & 3.3435 \end{pmatrix}$$

with mean vectors,

$$\mu_1 = (0.3, 0.25, 0.1), \mu_2 = (0.3, 0.25), \mu_3 = (0.3, 0.1)$$

constants $c_i's$ and parameters $a_i's$,

$$c = (1, 0.6, 0.4),\ a = (0.33, 0.33, 0.33)$$

By using the primal dual algorithm and setting the initial values for all the variables to be $x_0 = (\frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7}, \frac{1}{7})$, the asymptotic values $x_i's$ are,

$$x^* = (0.1515, 0.1136, 0.1587, 0.1516, 0.1138, 0.1520, 0.1590)$$

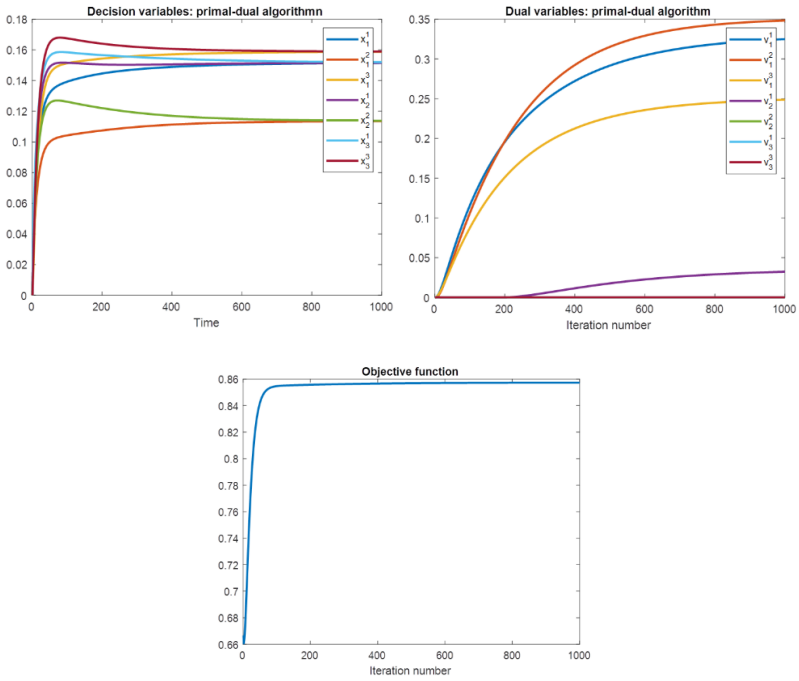


Fig. 2 (a) Portfolio Asset Allocation Variables (b) Dual Variables Convergence Process (c) Portfolio Risk Objective

with the total investment in each asset to be

$$x^{1*} = 0.1515 \times 3 = 0.4545$$

$$x^{2*} = 0.1136 \times 2 = 0.2272$$
$$x^{3*} = 0.1587 \times 2 = 0.3173$$

where the summation to 1 constraint is satisfied, $x^{1*} + x^{2*} + x^{3*} = 1$. The optimal value of the distributed optimization problem being

$$f(x^*) = 0.8573,$$

with each of the objective functions to be $f_1 = 1.2813, f_2 = 0.7281, f_3 = 0.5627$. We notice that our approach provides better optimal result and less risk for the investor. In figures (2) we present the convergence process of the primal variables (asset allocation), the dual variables (shadow prices) and the risk objective function. The asymptotic values of the dual variables are

$$v^* = (0.3249, 0.3484, 0.2486, 0.0322, 0, 0, 0)$$

For the pricing of the dual variables we will use the decentralized approach proposed in (6) and as a result, we have

$$p_i^{1*} = \left[\frac{v_i^{1*}}{v_1^{1*} + v_2^{1*} + v_3^{1*}} + 1\right] \cdot z_1, \ \forall \ i = 1, 2, 3.$$

$$p_i^{2*} = \left[\frac{v_i^{2*}}{v_1^{2*} + v_2^{2*}} + 1\right] \cdot z_2, \ \forall \ i = 1, 2.$$

$$p_i^{3*} = \left[\frac{v_i^{3*}}{v_1^{3*} + v_3^{3*}} + 1\right] \cdot z_3, \ \forall \ i = 1, 3.$$

where z_1, z_2 and z_3 denotes the standard pricing of the initial three assets x^1, x^2 and x^3 respectively. As a result we have,

$$p_1^{1*} = 1.91 z_1, p_2^{1*} = 1.09 z_1, p_3^{1*} = z_1$$

$$p_1^{2*} = 2 z_2, p_2^{2*} = z_2$$
$$p_1^{3*} = 2 z_3, p_3^{3*} = z_3.$$

3.3 Generalized Partitions Case

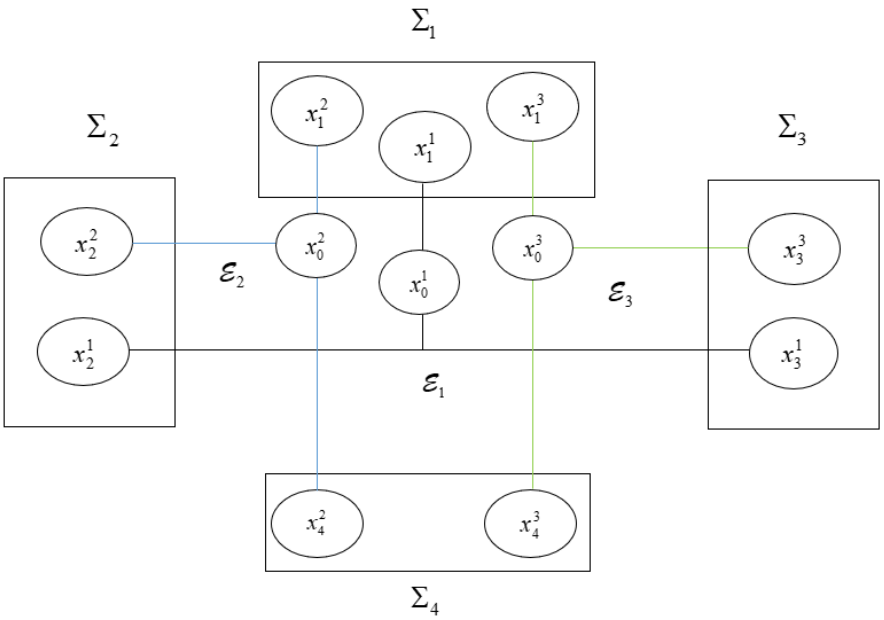


Fig. 3 Generalized Portfolio Overlapping

In this case we will use as sub-portfolios all the sets formed by the power set except from the empty set, i.e.,

$$\mathcal{P}(A) = \left\{\{x^1\}, \{x^2\}, \{x^3\}, \{x^1, x^2\}, \{x^1, x^3\}, \{x^2, x^3\}, \{x^1, x^2, x^3\}\right\}.$$

The respective mean variance distributed optimization problem would be of the form

$$\min_{[x_0, x_1, x_2, x_3, x_4]} \sum_{i=0}^{4} a_i f_i(x_i)$$

$$\begin{gathered} s.t. \quad x = Ez \\ x^T \cdot \mathbf{1} = 1, \ x_i^j \geq 0 \\ a^T \ \cdot \mathbf{1} = 1, \ a_i \geq 0 \end{gathered}$$

where

$$\begin{gathered} f_i(x_i) = x_i^T Q_i x_i - x_i^T \mu_i + c_i \\ x_0 = (x_0^1, x_0^2, x_0^3) \\ x_1 = (x_1^1, x_1^2, x_1^3) \\ x_2 = (x_2^1, x_2^2) \\ x_3 = (x_3^1, x_3^3) \\ x_4 = (x_4^2, x_4^3) \end{gathered}$$

with x_0 vector components denoting the actual assets in the case of no interactions among them. The variance-covariance matrices and the respective mean value vectors are provided below.

$$Q_0 = \begin{pmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \sigma_2^2 & 0 \\ 0 & 0 & \sigma_3^2 \end{pmatrix}, \mu_0 = (\mu_1, \mu_2, \mu_3)$$

$$Q_1 = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} & \sigma_{1,3} \\ \sigma_{2,1} & \sigma_2^2 & \sigma_{2,3} \\ \sigma_{3,1} & \sigma_{3,2} & \sigma_3^2 \end{pmatrix}, \mu_1 = (\mu_1, \mu_2, \mu_3)$$

$$Q_2 = \begin{pmatrix} \sigma_1^2 & \sigma_{1,2} \\ \sigma_{2,1} & \sigma_2^2 \end{pmatrix}, \mu_2 = (\mu_1, \mu_2)$$

$$Q_3 = \begin{pmatrix} \sigma_1^2 & \sigma_{1,3} \\ \sigma_{3,1} & \sigma_3^2 \end{pmatrix}, \mu_3 = (\mu_1, \mu_3).$$

$$Q_4 = \begin{pmatrix} \sigma_2^2 & \sigma_{2,3} \\ \sigma_{3,2} & \sigma_3^2 \end{pmatrix}, \mu_4 = (\mu_2, \mu_3).$$

The hypergraph incidence matrix E and the respective Laplacian matrix would be of the form:

$$E = \begin{pmatrix} 1&0&0\\ 0&1&0\\ 0&0&1\\ 1&0&0\\ 0&1&0\\ 0&0&1\\ 1&0&0\\ 0&1&0\\ 1&0&0\\ 0&0&1\\ 0&1&0\\ 0&0&1 \end{pmatrix}, L = \begin{pmatrix} \frac{3}{4}&0&0&\frac{-1}{4}&0&0&\frac{-1}{4}&0&\frac{-1}{4}&0&0&0\\ 0&\frac{3}{4}&0&0&\frac{-1}{4}&0&0&\frac{-1}{4}&0&0&\frac{-1}{4}&0\\ 0&0&\frac{3}{4}&0&0&\frac{-1}{4}&0&0&0&\frac{-1}{4}&0&\frac{-1}{4}\\ \frac{-1}{4}&0&0&\frac{3}{4}&0&0&\frac{-1}{4}&0&\frac{-1}{4}&0&0&0\\ 0&\frac{-1}{4}&0&0&\frac{3}{4}&0&0&\frac{-1}{4}&0&0&\frac{-1}{4}&0\\ 0&0&\frac{-1}{4}&0&0&\frac{3}{4}&0&0&0&\frac{-1}{4}&0&\frac{-1}{4}\\ \frac{-1}{4}&0&0&\frac{-1}{4}&0&0&\frac{3}{4}&0&\frac{-1}{4}&0&0&0\\ 0&\frac{-1}{4}&0&0&\frac{-1}{4}&0&0&\frac{3}{4}&0&0&\frac{-1}{4}&0\\ \frac{-1}{4}&0&0&\frac{-1}{4}&0&0&\frac{-1}{4}&0&\frac{3}{4}&0&0&0\\ 0&0&\frac{-1}{4}&0&0&\frac{-1}{4}&0&0&0&\frac{3}{4}&0&\frac{-1}{4}\\ 0&\frac{-1}{4}&0&0&\frac{-1}{4}&0&0&\frac{-1}{4}&0&0&\frac{3}{4}&0\\ 0&0&\frac{-1}{4}&0&0&\frac{-1}{4}&0&0&0&\frac{-1}{4}&0&\frac{3}{4} \end{pmatrix}$$

The variance-covariance matrices are,

$$Q_0 = \begin{pmatrix} 4.1294&0&0\\ 0&4.4150&0\\ 0&0&3.3435 \end{pmatrix}, Q_1 = \begin{pmatrix} 4.1294&1.4527&0.9273\\ 1.4527&4.4150&1.1068\\ 0.9273&1.1068&3.3435 \end{pmatrix}, Q_2 = \begin{pmatrix} 4.1294&1.4527\\ 1.4527&4.4150 \end{pmatrix},$$

$$Q_3 = \begin{pmatrix} 4.1294&0.9273\\ 0.9273&3.3435 \end{pmatrix}, Q_4 = \begin{pmatrix} 4.4150&1.1068\\ 1.1068&3.3435 \end{pmatrix}$$

with mean vectors,

$\mu^0 = (0.3, 0.25, 0.1), \mu^1 = (0.3, 0.25, 0.1), \mu^2 = (0.3, 0.25), \mu^3 = (0.3, 0.1), \mu^4 = (0.25, 0.1)$

constants $c_i's$ and parameters $a_i's$,

$$c = (0.7, 1, 0.6, 0.4, 0.9),\ a = (0.2, 0.2, 0.2, 0.2, 0.2).$$

By using the primal dual algorithm and setting the initial values for all the variables to be $x_i^j(0) = \frac{1}{12}\ \forall\ i, j$, the asymptotic values $x_i's$ are,

$x^* = (0.0893, 0.076, 0.0847, 0.0893, 0.076, 0.0847, 0.0893, 0.076, 0.0893, 0.0847, 0.076, 0.0847)$

with the total investment in each asset to be

$$x^{1*} = 0.0893 \times 4 = 0.3572$$

$$x^{2*} = 0.0760 \times 4 = 0.3040$$
$$x^{3*} = 0.0847 \times 4 = 0.3388$$

and the summation to 1 constraint to hold,

$$x^{1*} + x^{2*} + x^{3*} = 1.$$

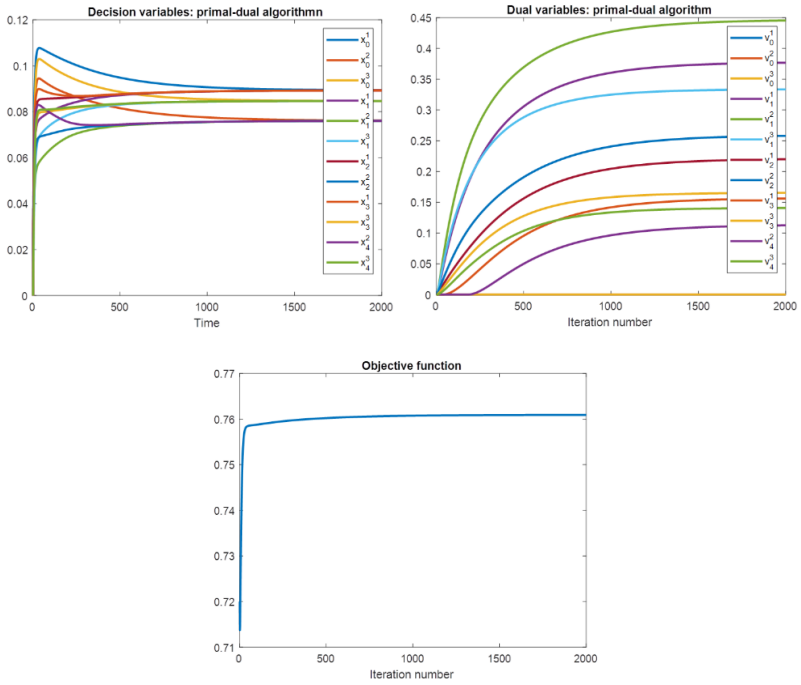


Fig. 4 (a) Portfolio Asset Allocation Variables (b) Dual Variables Convergence Process (c) Portfolio Risk Objective

with the optimal value of the distributed optimization problem being

$$f(x^*) = 0.7609,$$

with each of the objective functions to be $f_0 = 0.7282, f_1 = 1.0761, f_2 = 0.6324, f_3 = 0.4357, f_4 = 0.9323$. We notice that our approach provides better optimal result and less risk for the investor compared to the centralized approach. In figures (4) we present the convergence process of the primal variables (asset allocation), the dual variables (shadow prices) and the risk objective function. The asymptotic values of the dual variables are

$$v^* = (0, 0, 0, 0.3769, 0.4455, 0.3334, 0.2199, 0.2581, 0.1561, 0.1652, 0.1124, 0.1407)$$

For the pricing of the dual variables we will use the decentralized approach proposed in (6) and as a result, we have

$$p_i^{1*} = \left[\frac{v_i^{1*}}{v_0^{1*} + v_1^{1*} + v_2^{1*} + v_3^{1*}} + 1 \right] \cdot z_1, \ \forall\ i = 0, 1, 2, 3.$$
$$p_i^{2*} = \left[\frac{v_i^{2*}}{v_0^{2*} + v_1^{2*} + v_2^{2*} + v_4^{2*}} + 1 \right] \cdot z_2, \ \forall\ i = 0, 1, 2, 4.$$

$$p_i^{3*} = \left[\frac{v_i^{3*}}{v_0^{3*} + v_1^{3*} + v_3^{3*} + v_4^{3*}} + 1 \right] \cdot z_3, \ \forall \ i = 0, 1, 3, 4.$$

where z_1, z_2 and z_3 denote the standard pricing of the initial three assets x^1, x^2 and x^3 respectively. As a result we have,

$$p_0^{1*} = z_1, p_1^{1*} = 1.5z_1, p_2^{1*} = 1.29z_1, p_3^{1*} = 1.21z_1$$

$$p_0^{2*} = z_2, p_1^{2*} = 1.55z_2, p_2^{2*} = 1.32z_2, p_4^{2*} = 1.13z_2$$
$$p_0^{3*} = z_3, p_1^{3*} = 1.52z_3, p_3^{3*} = 1.26z_3, p_4^{3*} = 1.22z_3$$

The numerical results of this example clearly demonstrate the efficacy of our proposed decentralized portfolio framework in improving risk-adjusted outcomes through structured overlapping sub-portfolios. By transitioning from a centralized optimization problem to distributed formulations informed by hypergraph incidence structures, we enable a richer set of diversification opportunities while preserving feasibility and convexity. The observed reduction in the total risk objective and the enhanced interpretability of asset-specific pricing through dual variables reinforce the practicality of the approach. Moreover, the ability to model partial asset exposure using the powerset formulation allows for scalable generalizations to larger asset universes. These results support the potential of hypergraph-based decompositions in the design of modular, data-driven financial products with decentralized implementation and endogenous pricing.

4 Architectural Superiority of the Decentralized Hypergraph-Based Model

The results presented in the previous sections demonstrate that the decentralized portfolio optimization framework, which is built upon a hypergraph-based structure of overlapping sub-portfolios, offers a substantial advantage over traditional centralized approaches. This superiority arises not from any reformulation of the objective function or the imposition of complex regularization schemes, but from the architectural decentralization of the model itself.

4.1 Equivalence in Information of Problem Structure

Both the decentralized and centralized cases are informationally equivalent and aggregation-consistent, as both utilize the same input data-covariance matrices, expected returns and budget constraints, and allocate capital across the same asset space. Despite the use of distributed sub-portfolios and localized optimization problems, the decentralized model retains full informational parity with the centralized mean-variance formulation. Moreover, because the weights a_i on the local objectives are constrained to lie on the unit simplex (i.e., $\sum a_i = 1$), the decentralized formulation is informationally equivalent and aggregation-consistent with the centralized one in terms of overall optimization intent. The decentralized formulation acts as a structured decomposition of the centralized problem, where the solution process is distributed across overlapping sub-portfolios linked by a hypergraph. In particular,

each sub-portfolio problem $f_i(x_i)$ contributes linearly and proportionally to the global objective $\sum_i a_i f_i(x_i)$, and the hypergraph incidence structure ensures that shared assets are consistently reconciled across overlapping sub-portfolios via consensus constraints. While the variable structure and dimensionality differ, the solutions reside in the same feasible space, connected through a consensus constraint that ensures overall investment consistency. This structure enables a mathematically equivalent optimization landscape while structurally decomposing the computation. In this sense, the decentralized model can be viewed as a lifted representation that preserves the economic content of the centralized case while enabling enhanced diversification and modular risk control.

4.2 Decentralization Enables Systemic Diversification and Reduced Risk

Although the centralized and decentralized formulations are structurally isomorphic, the risk outcomes differ significantly. In the previous example the centralized solution yields an optimal risk of $f(x^*) = 2.8408$, while the decentralized approaches (both specific and powerset-based) yield significantly lower risk objectives, namely $f(x^*) = 0.8573$ and $f(x^*) = 0.7609$, respectively. These improvements occur without sacrificing total investment normalization or positivity constraints.

This result underscores the risk-reducing power of structured decentralization: by distributing decision-making across overlapping sub-portfolios and constraining them via hypergraph-induced consensus, the model achieves a form of systemic diversification. Each sub-problem manages a smaller, more manageable asset subset, often with simpler correlations, and the aggregated result is both globally consistent and more resilient to idiosyncratic risk.

4.3 Beyond Model Complexity: Robustness from Structure, Not Regularization

This section clearly distinguishes the proposed decentralized, hypergraph-based approach from the enhancements to classical portfolio optimization like those in Boyd et al. (2024). While techniques like shrinkage estimators, quadratic regularization, or second-order cone reformulations may enhance numerical performance, they are not central to the effectiveness of our framework. Instead, its strength lies in structural innovation—specifically, the decomposition of the portfolio into powerset-based sub-portfolios and the use of hypergraph couplings to distribute risk.

Operating under a deliberately simplified assumption space—quadratic objectives with linear constraints—the model leverages decentralization as a primary performance driver, rather than relying on parameter tuning. This architectural design enables robust diversification through modular risk-sharing and local interaction, maintaining strong performance.

Importantly, the framework can absorb and extend more complex formulations, including those of Boyd et al. (2024), without losing its structural advantage. Whether additional risk measures or regularization terms are introduced, the decentralized

structure continues to outperform a centralized implementation of the same enhancements, demonstrating robustness that stems from design rather than numerical complexity.

4.4 Decentralization Enables Interpretability and Pricing Mechanisms

An additional byproduct of the decentralized model is the natural emergence of localized dual variables, which are economically interpretable as *shadow prices* or *decentralized asset valuations*. These dual variables provide endogenous pricing mechanisms for assets, based on their relative marginal contributions to sub-portfolios, and allow for more context-sensitive valuation than global pricing in the centralized model.

As demonstrated in both the specific and generalized powerset cases, the dual variables enable a layered pricing scheme which reflects both global asset value and local importance within specific sub-portfolio configurations. This further emphasizes that pricing is no longer exogenous but emerges from the structural and functional decomposition of the problem.

4.5 Conclusion: Structural Superiority and Generalization Potential

This chapter has demonstrated that the decentralized hypergraph-based formulation yields superior portfolio performance not because of objective function complexity or data manipulation, but because of its structural innovation. The use of overlapping, interacting sub-portfolios mirrors real-world financial networks more accurately than traditional centralized optimizations, allowing for more granular risk control, better interpretability, and scalable generalizations.

Moreover, since the centralized and decentralized cases are isomorphic in informational content and objective structure, the performance gain must be attributed solely to the architectural benefits of decentralization. This offers a clear direction for future research and application: instead of optimizing more precisely within a centralized model, we should be optimizing more intelligently through decentralized structures.

5 Applied Example: GICS Sectors-Based ETF Allocation

To illustrate the practical applicability of the distributed portfolio optimization framework developed in the preceding sections, we turn to a concrete example using sector-based exchange-traded funds (ETFs) from the Global Industry Classification Standard (GICS). This setting naturally aligns with our model: GICS sectors represent economically meaningful clusters of assets, and their associated ETFs offer liquid, tradable instruments with well-defined risk-return profiles. By constructing overlapping sub-portfolios based on macroeconomic themes—each drawing from subsets of the GICS sectors set—we demonstrate how the theoretical concepts of hypergraph modeling, coupling constraints, and primal-dual optimization can be deployed in a

real-world investment context. The example not only validates the mathematical formulation but also showcases the flexibility and interpretability of our decentralized framework in capturing multi-theme/mandate investment strategies.

The Global Industry Classification Standard (GICS), developed by MSCI and Standard & Poor's, divides the equity universe into 11 major sectors. These sectors are widely used in investment analysis, index construction, and ETF product structuring. Each sector is associated with a liquid and widely-traded exchange-traded fund (ETF), as shown below:

GICS Sector	**Corresponding ETF (Ticker)**
Technology	XLK
Communication Services	XLC
Consumer Discretionary	XLY
Consumer Staples	XLP
Health Care	XLV
Utilities	XLU
Financials	XLF
Energy	XLE
Industrials	XLI
Real Estate	XLRE
Materials	XLB

These ETFs are well-suited for portfolio construction at the sector level, offering diversification, liquidity, and data availability.

5.1 Motivation for Using Sector-Based Sub-Portfolios

Rather than constructing a global portfolio across all 11 ETFs, we consider a more realistic and flexible setup: investors often design multiple investment plans based on macroeconomic views, thematic exposure, or sector rotation strategies. These "investment plans" are naturally modeled as **sub-portfolios** consisting of subsets from the power set of the 11 ETFs. We propose the use of 6 sub-portfolios that cover all sectors while reflecting practical investment behaviors. Each sub-portfolio corresponds to a thematically or risk-aligned strategy. The sub-portfolios and their sector compositions are:

Sub-Portfolio	**Constituent ETFs (Sectors)**
Σ_1	XLK, XLC, XLY, XLI, XLB (Growth + Innovation)
Σ_2	XLP, XLV, XLU, XLI (Defensive Core + Industrial Stability)
Σ_3	XLY, XLV, XLF, XLE, XLI (Cyclicals & Reflation Trade)
Σ_4	XLU, XLE, XLI, XLRE, XLB (Real Assets and Inflation Hedge)
Σ_5	XLK, XLC, XLY, XLP, XLV, XLI, XLRE (Balanced Growth & Core Exposure)
Σ_6	XLU, XLF, XLI, XLRE (Financials and Real Estate Core)

- Each sub-portfolio represents a distinct investment strategy, targeting specific economic themes or risk factors.
- The portfolios overlap in a way that allows for efficient risk sharing and coupling of asset allocation.

- All 11 sectors are represented across the sub-portfolios, ensuring broad market exposure.
- The structure enables investors to adjust their portfolio allocation according to evolving macroeconomic conditions, thus increasing flexibility.

Sub-Portfolio Interpretations

We now provide a detailed economic and financial rationale for the construction of six sub-portfolios based on GICS sector ETFs. Each sub-portfolio corresponds to a distinct thematic or macroeconomic view, often utilized by portfolio managers and institutional investors to gain targeted exposure or hedge specific risks. This methodology allows for a more adaptive approach to portfolio construction, making it possible to manage sector risk while maintaining exposure to core economic drivers.

Sub-Portfolio 1: Growth + Innovation

ETFs: XLK (Technology), XLC (Communication Services), XLY (Consumer Discretionary), XLB (Materials), XLI (Industrials)

This sub-portfolio targets sectors driven by high growth, innovation, and technological advancement, with the addition of materials and industrials to provide exposure to physical assets and infrastructure. It is well-suited for periods of expansion and technological progress.

Investor Profile: Growth-oriented investors with a focus on technological transformation, innovation, and infrastructure development.

Sub-Portfolio 2: Defensive Core + Industrial Stability

ETFs: XLP (Consumer Staples), XLV (Health Care), XLU (Utilities), XLI (Industrials)

This allocation targets defensive sectors, providing stability and income in times of market uncertainty, with the inclusion of industrials to add a degree of resilience and stability derived from physical assets. The combination of defensive sectors with industrial exposure ensures long-term stability with limited market sensitivity.

Investor Profile: Conservative investors seeking capital preservation, lower volatility, and consistent returns through sectors resilient in downturns.

Sub-Portfolio 3: Cyclicals & Reflation Trade

ETFs: XLF (Financials), XLE (Energy), XLI (Industrials), XLY (Consumer Discretionary), XLV (Health Care)

This portfolio focuses on cyclical sectors that benefit from economic recovery, reflationary environments, and policy-driven expansion. The inclusion of XLI (Industrials) enhances exposure to infrastructure and industrial growth, while XLY (Consumer Discretionary) reflects the positive impact of rising consumer sentiment.

Investor Profile: Investors positioning for economic recovery, increased growth, and fiscal stimulus, with exposure to sectors that perform well in expansionary periods.

Sub-Portfolio 4: Real Assets and Inflation Hedge

ETFs: XLB (Materials), XLE (Energy), XLRE (Real Estate), XLU (Utilities), XLI (Industrials)

This sub-portfolio targets sectors that provide protection against inflation through tangible assets. Materials and energy benefit from commodity price movements, while real estate and utilities provide shelter in inflationary environments. The industrials sector adds cyclical exposure that can perform well in both inflationary and growth periods.

Investor Profile: Investors seeking protection against inflation, with exposure to real assets and sectors that perform well during periods of rising prices.

Sub-Portfolio 5: Balanced Growth and Core Exposure

ETFs: XLK (Technology), XLC (Communication Services), XLY (Consumer Discretionary), XLP (Consumer Staples), XLI (Industrials), XLV (Health Care), XLRE (Real Estate)

This sub-portfolio provides a balanced exposure across growth (technology and communication services), defensive (consumer staples and health care), and cyclical (industrials) sectors, along with real estate for inflation protection. This mix allows for diversified exposure across varying economic cycles, with a focus on both growth and stability.

Investor Profile: Investors seeking a diversified, balanced approach to portfolio construction, with a blend of growth, defensive, and cyclical exposures.

Sub-Portfolio 6: Financials and Real Estate Core

ETFs: XLF (Financials), XLRE (Real Estate), XLI (Industrials), XLU (Utilities)

This sub-portfolio focuses on financials and real estate, with industrials and utilities providing additional economic exposure. Financials are a key sector during periods of economic recovery, while real estate offers inflation protection and cyclical upside. Utilities and industrials add stability and exposure to infrastructure.

Investor Profile: Investors seeking to capitalize on financial and real estate growth, with exposure to infrastructure and stability through utilities and industrials.

Summary Table

Sub-Portfolio	Theme	Key Characteristics
1	Growth + Innovation	High growth, tech-driven with physical asset sensitivity; includes technology, communication services, consumer discretionary, and industrials sectors.
2	Defensive Core + Industrial Stability	Recession-resilient, low-volatility with industrial exposure; combines consumer staples, health care, utilities, and industrials for stability and long-term growth.
3	Cyclicals & Reflation Trade	Positioned for economic recovery and policy-driven expansion; includes financials, energy, industrials, consumer discretionary, and health care for cyclical exposure.
4	Real Assets and Inflation Hedge	Protection against inflation through real assets; includes materials, energy, real estate, utilities, and industrials for exposure to inflation-sensitive sectors.
5	Balanced Growth and Core Exposure	Diversified across growth (tech), defensive (healthcare, staples), cyclical (industrials), and real estate sectors; suitable for varied economic conditions.
6	Financials and Real Estate Core	Focus on financials and real estate; includes industrials and utilities for infrastructure and stability, aiming for growth and protection in different economic cycles.

5.2 Optimization Formulation

For the centralized case we will use in the portfolio under consideration all 11 base ETFs. We will use actual data on a monthly basis from the period "2018-07-01" (inception of XLC ETF) till "2024-03-01", i.e., five years and 8 months. For the data collection we used the yahoo finance library in the python environment. The source code can be found in the Appendix of the paper. For the formulation of the optimization cases (centralized and decentralized) we will use the return mean vector and the variance covariance matrix which we extracted from the aforementioned data. In our setting we also include in our objectives fixed costs for choosing a specific type of portfolio. Incorporating sector ETFs into sub-portfolios incurs fixed implementation costs—such as transaction fees, research, and regulatory overhead—which vary by sector. We denote these as κ_j for sector j, and model total sub-portfolio cost as:

$$c_i = \kappa_i^\top \mu_i$$

This approach links costs to expected returns, reflecting that higher-return sectors often require greater complexity and expense to access. While an ideal cost model might use market data like bid-ask spreads, volatility, and expense ratios, data limitations lead us to a tiered heuristic approach based on sector characteristics:

- **High-cost**: Real Estate, Energy – $\kappa_j = 0.0025$
- **Medium-cost**: Industrials, Communication – $\kappa_j = 0.0020$
- **Low-cost**: Utilities, Staples – $\kappa_j = 0.0015$

Assigned Sector Costs

ETF (Sector)	Fixed Cost κ_j
XLK (Tech)	0.0025
XLC (Comm.)	0.0020
XLY (Disc.)	0.0020
XLP (Staples)	0.0015
XLV (Health)	0.0015
XLU (Utilities)	0.0012
XLF (Financials)	0.0018
XLE (Energy)	0.0022
XLI (Industrials)	0.0020
XLRE (Real Estate)	0.0021
XLB (Materials)	0.0020

These values reflect each sector's cost-efficiency and are scaled by return to account for implementation burden.

To ensure a more balanced and meaningful trade-off between risk and return, we adjust the portfolio budget constraint from $\sum x_i^j = 1$ to $\sum x_i^j = 1000$. Numerically, this rescaling addresses a common issue in quadratic optimization, where the quadratic risk term $x^T Q x$, typically on the order of 10^{-6}, becomes negligible compared to the linear return term $\mu^T x$, which is on the order of 10^{-3} when asset weights are constrained to sum to 1. This imbalance leads the optimization to focus almost entirely on maximizing return, often resulting in extreme, undiversified allocations. By increasing the portfolio scale, the quadratic term becomes more significant (scaling quadratically), restoring its influence in the objective function and promoting more realistic, diversified solutions. Financially, this can be interpreted as modeling an investor with a larger budget or considering positions in monetary units rather than normalized weights, making the model's outputs more interpretable and aligned with practical asset allocation scenarios.

5.2.1 Centralized Case

For the centralized case we will use all 11 ETFs in our portfolio, i.e., $A = \{x^1, ..., x^{11}\}$ where x^j is the jth ETF in order that we have presented them, e.g., x^1=XLK,..., x^{11}=XLB and we have the respective mean variance optimization problem to be of the form

$$\min_{x=(x^1,\ldots,x^{11})} f(x)$$
$$s.t.\ x^T \cdot \mathbf{1} = 1000,\ x^j \geq 0\ \forall\ j$$

where

$$f(x) = x^T Q x - x^T \mu + c$$

$$\mu = (0.019026, 0.009796, 0.010791, 0.008847, 0.010764, 0.007266, 0.010661, 0.012848,$$

$$0.011831, 0.007280, 0.010612)$$

$$Q = \begin{pmatrix} 0.004187 & 0.003338 & 0.004013 & 0.001720 & 0.001968 & 0.001505 & 0.002977 & 0.003223 & 0.003291 & 0.002686 & 0.003085 \\ 0.003338 & 0.003646 & 0.003438 & 0.001527 & 0.001699 & 0.001451 & 0.003102 & 0.003472 & 0.003004 & 0.002533 & 0.002938 \\ 0.004013 & 0.003438 & 0.004902 & 0.001706 & 0.002086 & 0.001666 & 0.003378 & 0.003811 & 0.003632 & 0.003238 & 0.003624 \\ 0.001720 & 0.001527 & 0.001706 & 0.001749 & 0.001435 & 0.001475 & 0.001916 & 0.002214 & 0.002103 & 0.001798 & 0.002025 \\ 0.001968 & 0.001699 & 0.002086 & 0.001435 & 0.002070 & 0.001299 & 0.002052 & 0.002500 & 0.002263 & 0.001842 & 0.002223 \\ 0.001505 & 0.001451 & 0.001666 & 0.001475 & 0.001299 & 0.002357 & 0.001653 & 0.001667 & 0.001836 & 0.002041 & 0.001941 \\ 0.002977 & 0.003102 & 0.003378 & 0.001916 & 0.002052 & 0.001653 & 0.004468 & 0.005418 & 0.003899 & 0.002769 & 0.003647 \\ 0.003223 & 0.003472 & 0.003811 & 0.002214 & 0.002500 & 0.001667 & 0.005418 & 0.011427 & 0.004748 & 0.002801 & 0.004486 \\ 0.003291 & 0.003004 & 0.003632 & 0.002103 & 0.002263 & 0.001836 & 0.003899 & 0.004748 & 0.004228 & 0.002868 & 0.003774 \\ 0.002686 & 0.002533 & 0.003238 & 0.001798 & 0.001842 & 0.002041 & 0.002769 & 0.002801 & 0.002868 & 0.003394 & 0.002934 \\ 0.003085 & 0.002938 & 0.003624 & 0.002025 & 0.002223 & 0.001941 & 0.003647 & 0.004486 & 0.003774 & 0.002934 & 0.004099 \end{pmatrix}$$

and

$$c = \mu^T \cdot \kappa = 0.0002155221.$$

We trivially find the asymptotic values,

$$x^{1*} = 0.00000458, x^{2*} = 17.17554023, x^{3*} = 0.00000333, x^{4*} = 490.29661206,$$

$$x^{5*} = 294.25683175, x^{6*} = 198.27099811, x^{7*} = 0.00000246, x^{8*} = 0.00000130,$$
$$x^{9*} = 0.00000156, x^{10*} = 0.00000291, x^{11*} = 0.00000165$$

with the optimal value of the centralized optimization problem to be

$$f(x^*) = 1589.5$$

For the pricing of the three assets standard methods can be used such as the CAPM model or the arbitrage pricing theory.

5.2.2 Decentralized Case

Let $z \in \mathbb{R}^{11}$ be the vector of investment weights or prices in the 11 base ETFs (corresponding to the GICS sectors). We have six sub-portfolios, each with its own decision variable x_i such that x_i^j denotes the allocation in the j-th ETF in i-th sub-portfolio. We have that the total number of variables to be $p = p_1 + ... + p_6 = 30$ and we define the incidence matrix of dimensions 30×11 by E which maps each coupling variable x_i^j to its corresponding ETF in z. As a result we have from the sub-portfolio incidence table below,

ETF to Sub-Portfolio Incidence Table

ETF (Sector)	Appears In Sub-Portfolios
XLK (Technology)	1, 5
XLC (Communication Services)	1, 5
XLY (Consumer Discretionary)	1, 3, 5
XLP (Consumer Staples)	2, 5
XLV (Health Care)	2, 3, 5
XLU (Utilities)	2, 4, 6
XLF (Financials)	3, 6
XLE (Energy)	3, 4
XLI (Industrials)	1, 2, 3, 4, 5, 6
XLRE (Real Estate)	4, 5, 6
XLB (Materials)	1, 4

the incidence matrix to be,

$$E = \begin{pmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
\hline
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0
\end{pmatrix}$$

from which we can calculate directly the hyperedge degree matrix

$$D_E = E^T E = \begin{pmatrix} 2&0&0&0&0&0&0&0&0&0&0&0 \\ 0&2&0&0&0&0&0&0&0&0&0&0 \\ 0&0&3&0&0&0&0&0&0&0&0&0 \\ 0&0&0&2&0&0&0&0&0&0&0&0 \\ 0&0&0&0&3&0&0&0&0&0&0&0 \\ 0&0&0&0&0&3&0&0&0&0&0&0 \\ 0&0&0&0&0&0&2&0&0&0&0&0 \\ 0&0&0&0&0&0&0&2&0&0&0&0 \\ 0&0&0&0&0&0&0&0&6&0&0&0 \\ 0&0&0&0&0&0&0&0&0&3&0&0 \\ 0&0&0&0&0&0&0&0&0&0&2&0 \end{pmatrix}.$$

We also have that $D_V = I_{30\times 30}$ and the Laplacian matrix which is of the form $L = D_V - ED_E^{-1}E^T = I - E(E^TE)^{-1}E^T$ is given by,

$$L = \begin{pmatrix} \frac{1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{2}&0&0&0&0&0&0&0&0&0 \\ 0&\frac{1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{2}&0&0&0&0&0&0&0&0 \\ 0&0&\frac{2}{3}&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0 \\ 0&0&0&\frac{5}{6}&0&0&0&0&\frac{-1}{6}&0&0&0&0&\frac{-1}{6}&0&0&\frac{-1}{6}&0&0&0&0&0&0&\frac{-1}{6}&0&0&0&\frac{-1}{6}&0 \\ 0&0&0&0&\frac{1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{2}&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&\frac{1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{2}&0&0&0&0&0&0 \\ 0&0&0&0&0&0&\frac{2}{3}&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&\frac{2}{3}&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0 \\ 0&0&0&\frac{-1}{6}&0&0&0&0&\frac{5}{6}&0&0&0&0&\frac{-1}{6}&0&0&\frac{-1}{6}&0&0&0&0&0&0&\frac{-1}{6}&0&0&0&\frac{-1}{6}&0 \\ 0&0&\frac{-1}{3}&0&0&0&0&0&0&\frac{2}{3}&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&\frac{-1}{3}&0&0&0&\frac{2}{3}&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&\frac{1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{2}&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&\frac{1}{2}&0&0&\frac{-1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&\frac{-1}{6}&0&0&0&0&\frac{-1}{6}&0&0&0&0&\frac{5}{6}&0&0&\frac{-1}{6}&0&0&0&0&0&0&\frac{-1}{6}&0&0&0&\frac{-1}{6}&0 \\ 0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&\frac{2}{3}&0&0&0&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{2}&0&0&\frac{1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&\frac{-1}{6}&0&0&0&0&\frac{-1}{6}&0&0&0&0&\frac{-1}{6}&0&0&\frac{5}{6}&0&0&0&0&0&0&\frac{-1}{6}&0&0&0&\frac{-1}{6}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{2}{3}&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&\frac{-1}{3} \\ 0&0&0&0&\frac{-1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{1}{2}&0&0&0&0&0&0&0&0&0&0 \\ \frac{-1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{1}{2}&0&0&0&0&0&0&0&0&0 \\ 0&\frac{-1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{1}{2}&0&0&0&0&0&0&0&0 \\ 0&0&\frac{-1}{3}&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0&0&0&0&0&\frac{2}{3}&0&0&0&0&0&0&0 \\ 0&0&0&0&0&\frac{-1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{1}{2}&0&0&0&0&0&0 \\ 0&0&0&0&0&0&\frac{-1}{3}&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0&0&0&0&0&0&\frac{2}{3}&0&0&0&0&0&0 \\ 0&0&0&\frac{-1}{6}&0&0&0&0&\frac{-1}{6}&0&0&0&0&\frac{-1}{6}&0&0&\frac{-1}{6}&0&0&0&0&0&0&0&0&\frac{5}{6}&0&0&0&\frac{-1}{6}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0&\frac{2}{3}&0&0&0&\frac{-1}{3} \\ 0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0&0&0&0&0&\frac{2}{3}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{2}&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{1}{2}&0&0 \\ 0&0&0&\frac{-1}{6}&0&0&0&0&\frac{-1}{6}&0&0&0&0&\frac{-1}{6}&0&0&\frac{-1}{6}&0&0&0&0&0&0&0&\frac{-1}{6}&0&0&0&\frac{5}{6}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{-1}{3}&0&0&0&0&0&0&0&0&0&0&\frac{2}{3} \end{pmatrix}$$

The Laplacian matrix enforces consensus (coupling) constraints across shared ETF exposures. Since L is an orthogonal projection matrix, we retain differentiability and convexity in the optimization dynamics. We see that the hypergraph structure is crucial for capturing the complex interdependencies between sectors that appear across multiple sub-portfolios. In traditional portfolio models, sectors are often treated in isolation, with each sector allocated independently. However, in our framework, some sectors appear in multiple sub-portfolios, reflecting real-world investment strategies

where certain sectors are strategically exposed to different macroeconomic views or market cycles. The hypergraph structure allows us to model these overlaps more effectively, treating each sector as a hyperedge that appears in multiple sub-portfolios. We formulate the decentralized mean-variance portfolio optimization problem as:

$$\min_{x=[x_1,\dots,x_6]} \sum_{i=1}^{6} a_i f_i(x_i)$$
$$\text{s.t. } x^T\mathbf{1} = 1000, \quad Lx = 0, \quad x \geq 0$$

with $x_i \in \mathbb{R}^{p_i}$ and $f_i(x_i) = x_i^T Q_i x_i - x_i^T \mu_i + c_i$. We choose the weights $a_i \in \mathbb{R}_+$ (risk preferences) of each sub-portfolio equal to $a_i = \frac{1}{6}$ and:

- for the first sub-portfolio we have the decision variable to correspond to the following ETFs
$$x_1 = (XLK, XLC, XLY, XLI, XLB)$$
with the respective mean vector, variance-covariance matrix to be:
$$\mu_1 = (0.019026, 0.009796, 0.010791, 0.011831, 0.010612)$$
$$Q_1 = \begin{pmatrix} 0.004187 & 0.003338 & 0.004013 & 0.003291 & 0.003085 \\ 0.003338 & 0.003646 & 0.003438 & 0.003004 & 0.002938 \\ 0.004013 & 0.003438 & 0.004902 & 0.003632 & 0.003624 \\ 0.003291 & 0.003004 & 0.003632 & 0.004228 & 0.003774 \\ 0.003085 & 0.002938 & 0.003624 & 0.003774 & 0.004099 \end{pmatrix}$$
and
$$c_1 = \mu_1^T \cdot \kappa_1 = 0.000127961$$
- for the second sub-portfolio we have the decision variable to correspond to the following ETFs
$$x_2 = (XLP, XLV, XLU, XLI)$$
with the respective mean vector, variance-covariance matrix to be:
$$\mu_2 = (0.008847, 0.010764, 0.007266, 0.011831)$$
$$Q_2 = \begin{pmatrix} 0.001749 & 0.001435 & 0.001475 & 0.002103 \\ 0.001435 & 0.002070 & 0.001299 & 0.002263 \\ 0.001475 & 0.001299 & 0.002357 & 0.001836 \\ 0.002103 & 0.002263 & 0.001836 & 0.004228 \end{pmatrix}$$
and
$$c_2 = \mu_2^T \cdot \kappa_2 = 0.0000855981$$
- for the third sub-portfolio we have the decision variable to correspond to the following ETFs
$$x_3 = (XLY, XLV, XLF, XLE, XLI)$$
with the respective mean vector, variance-covariance matrix to be:
$$\mu_3 = (0.010791, 0.010764, 0.010661, 0.012848, 0.011831)$$
$$Q_3 = \begin{pmatrix} 0.004902 & 0.002086 & 0.003378 & 0.003811 & 0.003632 \\ 0.002086 & 0.002070 & 0.002052 & 0.002500 & 0.002263 \\ 0.003378 & 0.002052 & 0.004468 & 0.005418 & 0.003899 \\ 0.003811 & 0.002500 & 0.005418 & 0.011427 & 0.004748 \\ 0.003632 & 0.002263 & 0.003899 & 0.004748 & 0.004228 \end{pmatrix}$$

and

$$c_3 = \mu_3^T \cdot \kappa_3 = 0.0000939106$$

- for the fourth sub-portfolio we have the decision variable to correspond to the following ETFs

$$x_4 = (XLU, XLE, XLI, XLRE, XLB)$$

with the respective mean vector, variance-covariance matrix to be:

$$\mu_4 = (0.007266, 0.012848, 0.011831, 0.007280, 0.010612)$$

$$Q_4 = \begin{pmatrix} 0.002589 & 0.002075 & 0.002274 & 0.001347 & 0.001959 \\ 0.002075 & 0.005291 & 0.002978 & 0.001348 & 0.002710 \\ 0.002274 & 0.002978 & 0.003778 & 0.001785 & 0.002894 \\ 0.001347 & 0.001348 & 0.001785 & 0.003089 & 0.001755 \\ 0.001959 & 0.002710 & 0.002894 & 0.001755 & 0.003929 \end{pmatrix}$$

and

$$c_4 = \mu_4^T \cdot \kappa_4 = 0.0000926946$$

- for the fifth sub-portfolio we have the decision variable to correspond to the following ETFs

$$x_5 = (XLK, XLC, XLY, XLP, XLV, XLI, XLRE)$$

with the respective mean vector, variance-covariance matrix to be:

$$\mu_5 = (0.019026, 0.009796, 0.010791, 0.008847, 0.010764, 0.011831, 0.007280)$$

$$Q_5 = \begin{pmatrix} 0.004187 & 0.003338 & 0.004013 & 0.001720 & 0.001968 & 0.003291 & 0.002686 \\ 0.003338 & 0.003646 & 0.003438 & 0.001527 & 0.001699 & 0.003004 & 0.002533 \\ 0.004013 & 0.003438 & 0.004902 & 0.001706 & 0.002086 & 0.003632 & 0.003238 \\ 0.001720 & 0.001527 & 0.001706 & 0.001749 & 0.001435 & 0.002103 & 0.001798 \\ 0.001968 & 0.001699 & 0.002086 & 0.001435 & 0.002070 & 0.002263 & 0.001842 \\ 0.003291 & 0.003004 & 0.003632 & 0.002103 & 0.002263 & 0.004228 & 0.002868 \\ 0.002686 & 0.002533 & 0.003238 & 0.001798 & 0.001842 & 0.002868 & 0.003394 \end{pmatrix}$$

and

$$c_5 = \mu_5^T \cdot \kappa_5 = 0.0001460645$$

- for the sixth sub-portfolio we have the decision variable to correspond to the following ETFs

$$x_6 = (XLU, XLF, XLI, XLRE)$$

with the respective mean vector, variance-covariance matrix to be:

$$\mu_6 = (0.007266, 0.010661, 0.011831, 0.007280)$$

$$Q_6 = \begin{pmatrix} 0.002357 & 0.001653 & 0.001836 & 0.002041 \\ 0.001653 & 0.004468 & 0.003899 & 0.002769 \\ 0.001836 & 0.003899 & 0.004228 & 0.002868 \\ 0.002041 & 0.002769 & 0.002868 & 0.003394 \end{pmatrix}$$

and

$$c_6 = \mu_6^T \cdot \kappa_6 = 0.0000714706$$

This setup allows us to compute a globally optimal decentralized investment strategy under local risk functions and shared asset constraints. By solving the distributed optimization problem we have:

$$x_1^* = (0, 42.1184, 0, 0, 50.0942)$$

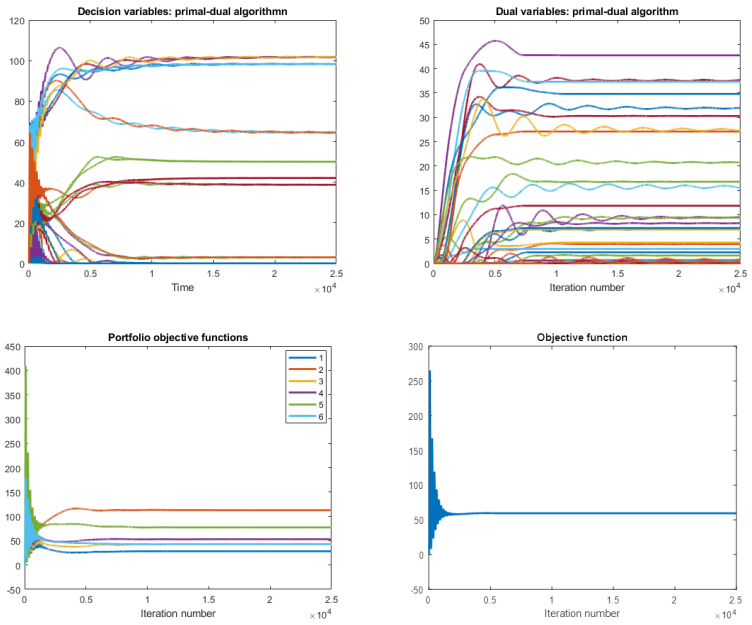


Fig. 5 (a) Portfolio Asset Allocation Variables (b) Dual Variables Convergence Process (c) Sub-portfolios Risk Objectives (d) Portfolio Risk Objective

$$x_2^* = (64.6181, 101.6761, 98.2671, 0)$$
$$x_3^* = (0, 101.7355, 38.7488, 0, 0)$$
$$x_4^* = (98.3239, 0, 0, 2.9548, 50.1075)$$
$$x_5^* = (0, 42.0994, 0, 64.4506, 101.7275, 0, 3.0275)$$
$$x_6^* = (98.3286, 38.7829, 0, 2.9383)$$

with the respective optimal sub-portfolio values to be

$$(f_1^*, f_2^*, f_3^*, f_4^*, f_5^*, f_6^*) = (28.2077, 112.6289, 43.1221, 53.0207, 77.2182,\ \ 42.8070)$$

and the total objective value to be

$$f(x^*) = 59.5007$$

which is significantly lower than 1589.5 which is the respective centralized optimal value. The pricing of the assets according the formula (5) can be achieved by using the asymptotic values of the dual variables which are given below:

$$v_1^* = (2.2672, 3.9460, 0.7152, 0.5966, 1.6087)$$

$$v_2^* = (15.5430, 37.6824, 31.9442, 27.0927)$$

$$v_3^* = (4.2758, 9.4762, 9.3726, 2.9742, 11.8359)$$

$$v_4^* = (7.0400, 0, 6.9564, 8.2674, 16.7755)$$

$$v_5^* = (37.3149, 30.2847, 34.8034, 0.8425, 27.2840, 42.7390, 20.7134)$$

$$v_6^* = (0.4213, 0.1858, 7.2547, 0.3018)$$

The convergence figures of the primal variables, dual variables, sub-portfolio values and total portfolio value are presented in Figure 5.

5.3 Exploring Sector ETF Subsets: Practical and Macroeconomic Perspectives

The 11 GICS sector ETFs yield a power set of $2^{11} = 2047$ (excluding the empty set) possible combinations. While only 6 sub-portfolios are used in this study for practical reasons, institutional investors such as hedge funds and investment banks could explore hundreds of these combinations. Their selection is guided by macroeconomic outlooks, risk preferences, and thematic exposures. This section presents a few illustrative subset examples—ranging from single-sector to multi-sector portfolios—to highlight the kind of economic reasoning that may inform these choices.

Single-Sector Subsets

- $\Sigma_{\text{Tech}} = \{\text{XLK}\}$ (Technology)
- $\Sigma_{\text{Health}} = \{\text{XLV}\}$ (Healthcare)

Rationale: Targeted bets on high-growth (Tech) or defensive (Healthcare) sectors.

Two-Sector Subsets

- $\Sigma = \{\text{XLK}, \text{XLY}\}$ (Tech + Consumer Discretionary)
- $\Sigma = \{\text{XLV}, \text{XLP}\}$ (Healthcare + Staples)

Rationale: Balanced exposure to growth and consumer demand, or a fully defensive tilt.

Four-Sector Subsets

- $\Sigma = \{\text{XLK}, \text{XLF}, \text{XLI}, \text{XLB}\}$ (Growth + Cyclicals)
- $\Sigma = \{\text{XLV}, \text{XLP}, \text{XLU}, \text{XLRE}\}$ (Defensive sectors)

Rationale: Broader diversification across either economically sensitive or low-volatility sectors.

Seven-Sector Subsets

- $\Sigma = \{\text{XLK}, \text{XLY}, \text{XLV}, \text{XLF}, \text{XLI}, \text{XLB}, \text{XLE}\}$
- $\Sigma = \{\text{XLF}, \text{XLV}, \text{XLU}, \text{XLP}, \text{XLRE}, \text{XLE}, \text{XLB}\}$

Rationale: Strategic allocation with tilt toward either recovery-driven growth or stability and income.

Nine-Sector Subsets

- $\Sigma = \{\text{XLK}, \text{XLY}, \text{XLF}, \text{XLV}, \text{XLI}, \text{XLB}, \text{XLRE}, \text{XLU}, \text{XLE}\}$
- $\Sigma = \{\text{XLK}, \text{XLV}, \text{XLF}, \text{XLP}, \text{XLU}, \text{XLRE}, \text{XLB}, \text{XLI}, \text{XLC}\}$

Rationale: Broad market representation with macro-hedging built in via sectoral diversification.

These examples illustrate how different sector combinations reflect varying macroeconomic perspectives and strategic intents. While our study limits itself to six sub-portfolios for simplicity, real-world applications often involve much larger selections—each one tailored to specific return targets, risk appetites, and economic narratives.

5.4 Future Work

While the proposed methodology for generating customized portfolios through sub-portfolios shows promising results, several potential extensions and refinements could further enhance its applicability and robustness. These future directions could include, but are not limited to, the following:

- **Smart Subset Selection via Machine Learning:** Instead of relying solely on arbitrary or exhaustive subset selection from the power set, incorporating machine learning techniques such as clustering, principal component analysis (PCA), or even reinforcement learning could help identify informative or non-redundant subsets. This would not only improve computational efficiency but also uncover latent portfolio structures that may not be immediately apparent.
- **Dynamic Rebalancing in Sub-Portfolios:** A dynamic extension of the current methodology could be introduced, where sub-portfolios are rebalanced over time to adapt to changing correlations, market volatilities, or macroeconomic signals. This could be modeled using differential equations or control theory to maintain optimal portfolio alignment as market conditions evolve.
- **Behavioral Finance Angle:** The concept of sub-portfolios could be extended to cater to investor psychology and risk profiles. By segmenting portfolios into different characteristics (e.g., aggressive, defensive, ESG-heavy), the methodology could serve as a tool to design portfolios aligned with behavioral finance principles, providing a more personalized investment solution.
- **Tokenization of Sub-Portfolios:** A futuristic extension of the proposed framework could involve tokenizing each sub-portfolio or creating blockchain-based financial products. This would enable the sub-portfolios to be traded on digital

asset platforms, offering liquidity and access to a broader range of investors in the growing space of decentralized finance.

By exploring these avenues, the methodology could evolve into a more flexible, scalable, and adaptive framework, increasing its utility across a wide range of investment strategies and market conditions.

6 Conclusion

This paper presents a novel distributed portfolio optimization framework based on hypergraph structures and combinatorial decomposition. By organizing asset allocations through overlapping sub-portfolios—drawn from the power set of available assets—we model realistic investment architectures such as thematic strategies, multi-manager funds, and pension mandates. The primal-dual formulation enables consistent allocations across sub-portfolios while introducing shadow prices for coupling assets, offering a natural economic interpretation rooted in general equilibrium. Our approach allows for integration of external pricing models (e.g., CAPM) and internal constraints, creating a unified optimization mechanism. Using GICS sector ETFs as a practical case, we demonstrate how macroeconomic sub-portfolios can be constructed efficiently, yielding improved risk-adjusted outcomes. The method generalizes to broader settings and provides a scalable, interpretable tool for decentralized asset management.

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7 Appendix

7.1 Non Linear Control Theory Preliminaries for the Proof of the Theorem

We study the following continuous, time invariant system,

$$\dot{x}(t) = f(x(t)) \tag{7}$$

with $x(t) \in \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}^n$ being continuous. A function $V : \mathbb{R}^n \to \mathbb{R}$ for which the following relationship $||x|| \to \infty \Rightarrow V(x) \to \infty$ is satisfied is called radially unbounded. By $\dot{V} : \mathbb{R}^n \to \mathbb{R}$ is denoted the Lie derivative of V which is expressed by the following formula:

$$\dot{V}(x(t)) = \nabla V(x(t))^T \cdot \dot{x}(t) = \nabla V(x(t))^T \cdot f(x(t)).$$

Theorem 2. *Let x^* be an equilibrium point of (7). If $V : \mathbb{R}^n \to \mathbb{R}$ is positive definite, radially unbounded and $\dot{V}(x) < 0, \forall\ x \neq x^*$ then x^* is globally asymptotically stable and V is a valid Lyapunov function for (7).*

Definition 1. *A domain $\mathcal{D} \subseteq \mathbb{R}^n$ is called invariant for (7) if*

$$\forall\ x(t_0) \in D \Rightarrow x(t) \in D,\ \forall\ t \in \mathbb{R}.$$

Definition 2. *A domain $\mathcal{D} \subseteq \mathbb{R}^n$ is called positively invariant for (7), if*

$$\forall\ x(t_0) \in D \Rightarrow x(t) \in D,\ \forall\ t \geq t_0.$$

Theorem 3 (LaSalle's Invariance Principle)**.** *Let $\Omega \subset D$ be a compact possitively invariant set with respect to (7). Let $V : D \to \mathbb{R}$ be a continuously differentiable function such that $\dot{V}(x) \leq 0$ in Ω. Let $\mathcal{X}$ be the set of all points in Ω where $\dot{V}(x) = 0$. Let M be the largest invariant set in $\mathcal{X}$. Then every solution starting in Ω approaches M as $t \to \infty$.*

Lemma 4. *Considering a convex function $f : \mathbb{R}^n \to \mathbb{R}^n$, its gradient $\nabla f : \mathbb{R}^n \to \mathbb{R}^n$ is incrementally passive, i.e., the following inequality holds ,*

$$(\nabla f(x) - \nabla f(y))^T (x - y) \geq 0,\ \forall\ x, y \in \mathbb{R}^n. \tag{8}$$

If f is strictly convex, inequality (8) strictly holds $\forall x \neq y$ and then ∇f is strictly incrementally passive.

7.2 Lemma

The following lemma is used in the proof of the main convergence theorem.

Lemma 5. *The hypergraph Laplacian matrix L in (3) is an orthogonal projection matrix.*

Proof. We have that

$$\begin{aligned}
L^2 =&[I - E(E^TE)^{-1}E^T]^2 \\
=&[I - E(E^TE)^{-1}E^T][I - E(E^TE)^{-1}E^T] \\
=&I - 2E(E^TE)^{-1}E^T + E(E^TE)^{-1}E^TE(E^TE)^{-1}E^T \\
=&I - 2E(E^TE)^{-1}E^T + E(E^TE)^{-1}E^T \\
=&I - E(E^TE)^{-1}E^T \\
=&Q
\end{aligned}$$

and also $L = L^T$. As a result, L is an orthogonal projection matrix. Matrix L is also positive semidefinite with spectrum $S(L) = \{0, 1\}$. □

7.3 Proof of Theorem 2

Proof. We define the auxiliary variables:

$$g(x, \lambda, v) := \nabla f(x) - Lv - \lambda \mathbf{1}, \quad r(x) := \mathbf{1}^\top x - 1, \quad s(x) := Lx.$$

and we introduce the Lyapunov function:

$$V(x, \lambda, v) := \frac{1}{2}\|g(x, \lambda, v)\|^2 + \frac{1}{2}r(x)^2 + \frac{1}{2}\|s(x)\|^2.$$

This function is non-negative and equals zero if and only if the KKT conditions are satisfied. Differentiating V with respect to time:

$$\dot{V} = \langle g, \dot{g} \rangle + r\dot{r} + \langle s, \dot{s} \rangle.$$

We compute:

$$\begin{aligned}
\dot{g} &= \nabla^2 f(x)\dot{x} - L\dot{v} - \dot{\lambda}\mathbf{1}, \\
\dot{r} &= -\mathbf{1}^\top g, \\
\dot{s} &= -Lg.
\end{aligned}$$

Substituting $\dot{x} = -g$, $\dot{v} = -s = -Lx$, $\dot{\lambda} = -r$:

$$\begin{aligned}
\langle g, \dot{g} \rangle &= -g^\top \nabla^2 f(x) g + g^\top L^2 x + r g^\top \mathbf{1}, \\
r\dot{r} &= -r\mathbf{1}^\top g, \\
\langle s, \dot{s} \rangle &= -s^\top g.
\end{aligned}$$

Using $L = L^\top = L^2$ and $s = Lx$:

$$\dot{V} = -g^\top \nabla^2 f(x) g + g^\top Lx + r g^\top \mathbf{1} - r\mathbf{1}^\top g - g^\top Lx = -g^\top \nabla^2 f(x) g \leq 0.$$

Since $f(x)$ is strongly convex, $\nabla^2 f(x) \succ 0$, hence $\dot{V} = 0$ if and only if $g = 0$, $r = 0$, and $s = 0$. This implies:

$$\nabla f(x^*) = Lv^* + \lambda^* \mathbf{1}, \quad \mathbf{1}^\top x^* = 1, \quad Lx^* = 0,$$

i.e., the KKT conditions are satisfied. To handle the non-negativity constraint $x \geq 0$, since the vector field is Lipschitz and the constraint set is convex and closed, standard invariance principles imply that any limit point of the trajectory satisfies $x \geq 0$. By LaSalle's Invariance Principle, the solution trajectory $(x(t), \lambda(t), v(t))$ converges to the largest invariant set in $\{\dot{V} = 0\}$, which consists only of points satisfying the KKT conditions. Therefore, the system globally converges to an optimal solution. □

7.4 Python Code for ETF Return Analysis

```
import yfinance as yf
import pandas as pd

# Define the ETF tickers in the specified order
tickers = ['XLK', 'XLC', 'XLY', 'XLP', 'XLV', 'XLU', 'XLF', 'XLE', 'XLI', 'XLRE', 'XLB']

# Download monthly adjusted close prices for the following timeframe
data = yf.download(tickers, start='2018-07-01', end='2024-03-01', interval='1mo')['Adj Close']
data.dropna(inplace=True)

# Calculate monthly returns
returns = data.pct_change().dropna()

# Mean vector and covariance matrix
mean_vector = returns.mean()
cov_matrix = returns.cov()

# Convert to LaTeX
mean_vector_latex = mean_vector.to_frame(name='Mean Return').T.to_latex(float_format="%.6f")
cov_matrix_latex = cov_matrix.to_latex(float_format="%.6f")
returns_latex = returns.to_latex(float_format="%.6f", longtable=True)

# Save to files if needed
with open("mean_vector.tex", "w") as f:
f.write(mean_vector_latex)

with open("cov_matrix.tex", "w") as f:
f.write(cov_matrix_latex)
```

```
29
30          with open("returns_data.tex", "w") as f:
31          f.write(returns_latex)
```

Listing 1 Download Adjusted Prices, Compute Returns, Mean Vector, and Covariance Matrix